Figure 1: Self-IoT-AD Dual-Stage Benchmark Architecture & Representation Learning Pipeline

Leakage-Free Feature Deduplication • Corrupt-Contrastive Tabular Pretext Engine • Momentum Teacher Consistency Evaluation

RT-IoT2022 Curated

PyTorch 2.14 Verified

ZONE 1: TELEMETRY & AUDIT
Raw Ingestion & Leakage Elimination

Physical IoT Testbed
- RT-IoT2022 Network Monitors
- ThingSpeak, MQTT, Smart Bulbs
- 21 Continuous Flow Metrics

Leakage-Free Curation
- Exact duplicate audit (1-NN d=0)
- Raw: 76.06% test duplicate leakage
- Curated: 17,710 unique flows (0.45%)

ZONE 2: TABULAR PRETEXT PRETRAINING
Corrupt-Contrastive SSL on Unlabeled Flows

Marginal Permutation
- Feature-wise marginal sampling
- Bernoulli mask m ~ Bern(0.30)
- Preserves univariate marginals

Dual Pretext Engine
- Neural Encoder: 21 -> 128 -> 64
- InfoNCE contrastive mutual info
- Denoising decoder MSE recon

ZONE 3: CONSISTENCY FINE-TUNING
Semi-Supervised Downstream Adaptation

Pretrained Weights

Scarce Labeled Telemetry
- 1% budget: 123 labeled flows
- 5% budget: 619 labeled flows
- Class-balanced focal loss (g=2)

Momentum Teacher Loop
- EMA student tracking (alpha=0.996)
- Confidence-gated soft targets
- Ablation: Teacher harms tabular F1

ZONE 4: MULTI-MODEL BENCHMARK
11 Architectures & Linear Probing Suite

Linear Probe Benchmark
- Frozen pretext representation Z
- Logistic probe across 20 seeds
- F1 = 87.59% (p = 0.0042 vs end-to-end)

11-Architecture Evaluation
- Tree Ensembles: RF (93.5%), XGB (92.1%)
- Tabular SSL: VIME (91.6%), SCARF (91.2%)
- Unsupervised: IsoForest, OCSVM, DAE